Hasan Mohd Hussain

Data Engineer | ETL/ELT Data Pipelines | SQL, Python, Spark & AWS | Data Modeling
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PROFESSIONAL SUMMARY

Data Engineer with an **MS in Computer Engineering**, building **ETL/ELT data pipelines**, **API data integration** from **145+ sources** and **data quality** automation in **Python** and **SQL**. Shipped a **RAG ingestion pipeline** with **30%** better retrieval.

EDUCATION

University of Texas at Dallas — **Richardson, Texas, USA**
Master of Science in Computer Engineering — Aug 2024 - Aug 2026
GPA: 3.74/4.0 | *Relevant Coursework: NLP, Machine Learning, Computer Architecture, Embedded Systems*

University of Mumbai — **Mumbai, Maharashtra, India**
Bachelor of Engineering in Computer Engineering — Aug 2017 - Jul 2021
GPA: 7.88/10.0 | *Relevant Coursework: Data Structures, Database Management Systems*

SKILLS

Languages: SQL (joins, CTEs, window functions), Python, PL/SQL, Scala, Java, C++, Bash, shell scripting
Data Engineering: ETL/ELT, data pipelines, data modeling (star schema, SCD), data warehousing, data lake design, data integration, transformation, validation, CDC, batch & streaming, data lineage, data governance, documentation, Amazon Redshift, BigQuery, distributed systems
Big Data & Orchestration: Apache Spark (PySpark), Databricks, Delta Lake, Apache Airflow, Apache Kafka, Amazon MSK, dbt, notebooks
Databases & Warehousing: Snowflake, PostgreSQL, Oracle, MySQL, SQLite, MongoDB, ChromaDB, indexing, query optimization
Cloud & DevOps: AWS (Glue, S3, Lambda, EC2, ECS Fargate, ECR), Docker, Kubernetes, CI/CD, Git, Linux, Datadog, monitoring, logging
Analytics & Quality: Pandas, NumPy, data quality, Great Expectations, PyTest (unit testing), FastAPI, REST APIs, Oracle Data Integrator (ODI), Oracle Analytics Cloud (OAC), Power BI, Tableau, Amazon Athena, Excel
AI Tools: OpenAI API, Azure OpenAI, Ollama, LangChain, RAG, vector embeddings, Claude Code

WORK EXPERIENCE

Mphasis — **New York, United States**
Machine Learning Engineer — Jun 2026 - Aug 2026
- Engineered an **automated QA and data validation agent** in **Python, Flask and Playwright** for a **production mortgage dashboard**.
- Automated **LLM-as-a-judge evaluation** across **Claude, GPT-4o, Gemini and Qwen** with **PDF/Excel reporting**, cutting test time **95%**.
- Developed a **synthetic data generation pipeline** with answer keys for **5 financial document types**, producing **200 eval cases/run**.

Mphasis — **New York, United States**
Machine Learning Engineer — May 2025 - Aug 2025
- Delivered a **RAG ingestion pipeline** (**PDF extraction, chunking, embeddings**) into **ChromaDB** with **FastAPI**, improving retrieval **30%**.
- Transformed **JSON bookmark metadata** with **Python/PyPDF2** to automate **loan-document data processing** (page reordering, rotation).
- Optimized **Python backend logic** for **data-serving REST APIs**, improving **API response time by 20%**.

Dot9 Games — **Mumbai, India**
Game Developer Intern — Apr 2024 - Aug 2024
- Built **Unity/C#** gameplay systems from design docs, raising **user engagement 25%** at a stable **60 FPS**, using **Git**.

Shree Infotech — **Mumbai, India**
Software Engineer — Apr 2023 - Mar 2024
- Developed **Python/Flask backend services** and **REST APIs** on **Docker**, automating workflows and saving **10+ hours/week**.
- Designed **MongoDB** schemas and indexes for **fast query retrieval**, and wrote **unit tests** (PyTest) to keep core services reliable.

Nilee Games — **Mumbai, India**
Game Developer Intern — Nov 2022 - Mar 2023
- Shipped a **2D physics-based driving game** in **Unity/C#** in **two weeks**: **500K downloads** in month one, **30%** higher engagement.

PROJECTS

TaxiStream Lakehouse | *PySpark, Delta Lake, Kafka (Redpanda), Airflow, dbt, Great Expectations*
- Architected a **gold star schema** (**1 fact, 5 dimensions**) on **bronze/silver/gold Delta Lake** for a planned **9.5M-trip taxi** backfill.
- Building **PySpark** batch and **Redpanda (Kafka)** streaming ingestion with **Airflow**-run **dbt** marts behind **3 Great Expectations** gates.

JobQuest.AI | *Python, FastAPI, SQLite, REST APIs, ETL, Data Pipelines*
- Implemented an **ETL pipeline** ingesting postings from **145+ job boards** via **Greenhouse, Lever, Ashby** and **Workday REST APIs**.
- **Normalized and deduplicated** records into a **12-table SQLite** schema powering **Chart.js dashboards**, backed by **195 automated tests**.

AI SQL Data Analyst Agent | *Python, SQL, LangChain, Gemini, SQLite, SQLAlchemy*
- Built a **text-to-SQL agent** in **LangChain** that generates and executes **multi-table SQL joins** on the **11-table Chinook database**.
- Surfaced **query results and data insights** for **natural-language analytics questions** over **15,607 rows** via an interactive **CLI**.

CERTIFICATIONS

Anthropic: Claude Code in Action, Claude Code 101, Claude 101 | **Udemy:** SQL Bootcamp | **DeepLearning.AI:** AI For Everyone
IBM: Machine Learning with Python, Building AI Agents and Agentic Workflows | **Meta:** Back-End Developer Professional Certificate